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Solar cells: past, present, future

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Abstract

This paper gives an up to date perspective of PV technology and materials. The most important material has been and still is silicon. It dominates the present world market, particularly in its crystalline form but amorphous silicon is also of importance. Crystalline silicon solar cells are still heavily dependent on the materials base of the semiconductor industry. This material still has a large potential for cost reduction in its conventional form and even more so in the crystalline thin film version. Great hope rests with the thin film materials which require only small amounts of material. A great number of new concepts and materials are still in the research stage. Some of them may lead to much higher efficiency and lower cost in the coming decades. A function was identified which can be fitted to the past development of the best laboratory efficiencies. By fitting this function to the data points the future development of efficiency can be estimated for each technology. © 2002 Elsevier Science B.V. All rights reserved.

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1. Photovoltaic effect and principle of solar cell operation

The conversion of solar radiation occurs by the photovoltaic effect which was first observed by Becquerel [1]. It is quite generally defined as the emergence of an electric voltage between two electrodes attached to a solid or liquid system upon shining light onto this system. Practically, all photovoltaic devices incorporate a pn junction in a semiconductor across which the photovoltage is developed.

A large part of the paper will be concerned with silicon although from solid state physics we know that silicon is not the ideal material for photovoltaic conversion. The requirements for the ideal solar cell material follow from solid state physics. For

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a 90% light absorption it takes only 1 μm of GaAs (a direct semiconductor) versus 100 μm of crystalline Si. The photogenerated carriers have to reach the pn junction which is mostly near the front surface. The diffusion length of minority carriers for high efficiency has to be 200 μm or at least twice the silicon thickness. Thus the material has to be of very high purity and of high crystalline perfection. In view of these physical limitations it is quite surprising that silicon has played such a dominant role in the market. The main reason is that silicon technology has already been highly developed before the advent of photovoltaics and high-quality material is being produced in large quantities for the microelectronics market.

It is not surprising, on the other hand, that a lot of effort has been going on and is still going into the search for new materials. Requirements for the ideal solar cell material are: (1) band gap between 1.1 and 1.7 eV (2); direct band structure (3); consisting of readily available, non-toxic materials; (4) easy, reproducible deposition technique, suitable for large area production; (5) good photovoltaic conversion efficiency; and (6) long-term stability. For the future of solar energy materials three scenarios can be envisioned:

- Continued dominance of the present single crystal or cast polycrystal technology.
- New crystalline film Si materials of medium thickness either as ribbons or on foreign substrates.
- Breakthrough of true thin film materials like a-Si, CIS or CdTe.

At this point those scenarios have about equal probability. Even more likely is that two or three of them will coexist for a considerable period and that each technology will find its own market. In the long term new concepts or new classes of materials like organic solar cells also are a possibility. They will be addressed towards the end of this paper. From an overall point of view, it can be considered an advantage that so many avenues exist that potentially lead to a low-cost high-efficiency solar cell. In this way the likelihood of achieving this goal is greatly increased.

2. Monocrystalline and multicrystalline silicon

2.1. History

The first silicon solar cell was developed at Bell Laboratories in 1954 by Chapin et al. [2]. It already had an efficiency of 6% which was rapidly increased to 10%. The main application for many years was in space vehicle power supplies.

2.1.1. Status today

Today's photovoltaic market is characterized by the following trends:

- Slow but steady improvement of conversion efficiency.
- Slow reduction of the cost of modules and systems.
- Uncertain supply base of polycrystalline raw material.

In this context the importance of conversion efficiency has to be discussed. Except for some applications in consumer products efficiency is very important because of area related cost which constitutes a large part of systems cost. Therefore, both in research laboratories and in manufacturing, improvement of efficiency is of high priority. The best laboratory efficiency for single crystal silicon is today 24.7% [3]. This efficiency can only be realized with very elaborate technology but experience has shown that progress in laboratory efficiency leads to corresponding improvement in production with a certain time delay.

2.1.2. *The silicon supply problem*

A big question mark for the future is the source of highly purified silicon for solar cells. Fifty percent of the cost of a module is due to the cost of processed silicon wafers. The PV industry has in the past used reject material from the semiconductor industry that was available at low cost. This created a dependence that is only viable if both sectors grow at the same rate. An additional problem is that the semiconductor market is characterized by violent cycles of boom and depression superimposed on a relatively steep growth curve. In boom times the materials supply becomes tight and prices increase. This happened in 1998 when even reject material was in short supply and some solar cell manufacturers had to buy regular semiconductor grade material at relatively high cost. Today silicon supply is again very critical.

2.1.3. *Future*

The present technology is relatively mature but several studies have shown that it still has a large cost reduction potential. As for other industrial products, manufacturing cost follows a learning curve which indicates that for every doubling of manufacturing volume, cost drops by 20%. With this learning curve for conventional technology it would take a relatively long time to reach a significant contribution of PV to world energy supply. On the other hand, the crystalline technology has proven itself in the past to be a moving target. Amorphous silicon, for instance, has not been able to extend its market share. The only major new development required is a source of solar grade silicon as was outlined above. It will probably become available when demand is large enough.

3. The main approaches in ribbon silicon production

In the *edge-defined film-fed growth process* utilized by ASE Americas, a self-supporting silicon ribbon is pulled from the melt through a die which determines the shape of the ribbon [4]. Today octagon tubes of 5.3 m length at a nominal average wall thickness of 280 μm are pulled out of a graphite crucible containing liquid silicon and are subsequently separated by a laser. The resulting sheets of $10 \times 10 \text{ cm}^2$ have a somewhat lower material quality than single crystals, and they have a wavy surface. Nevertheless, conversion efficiencies of up to 14.8% were achieved in the production line with an excellent overall yield of over 90% at the moment [5].

The *string ribbon process* is under development since 1994 at the Evergreen company and the principles are described in Ref. [6]. Silicon ribbons of variable thicknesses are pulled with two temperature resistant strings directly out of the melt and are cut subsequently into the desired length with diamond tools. The growth speed is up to 25 mm/min resulting in ribbons with a thickness of below 100 μm and a conversion efficiency of 15.1% on a lab cell of 1 cm^2 area. There are 30 and 60 W modules commercially available consisting of cells of $5.6 \times 15 \times 0.025 \text{ cm}^3$ area [7].

4. Crystalline thin-film silicon

4.1. History

The first considerations concerning thinner silicon wafers for solar cells were made by Wolf [8] and Lofersky [9], while simulating the ideal parameters for record high efficiencies. They pointed out that with decreasing cell thickness the open circuit voltage increases due to the reduced saturation current which again is a result of a decreasing geometry factor. The first theoretical work about light trapping in silicon layers suggested a lambertian back reflector as a simple but efficient structure [10].

4.2. The basic components of a crystalline silicon thin-film solar cell (c-SiTFE)

The linking feature of all c-SiTFE approaches is the underlying substrate needed as a mechanical support due to the reduced thickness of the active silicon layer of typically 5–50 μm . The substrate consists either of low quality silicon or of foreign materials such as glass, ceramics or graphite. The choice of the substrate material determines the maximum allowed temperature for solar cell processing and therefore nearly all c-SiTFE approaches can be assigned to one of the three categories: (1) high-temperature approach; (2) low-temperature approach; and (3) transfer techniques, which are described below.

4.3. The present status of the crystalline silicon thin film solar cell

4.3.1. Si-layers deposited directly onto glass

The inherent limitation of all approaches compatible with glass substrates is the softening point of the glass which is in the order of 600°C. This means that both the Si-deposition and the solar cell processing are not allowed to go beyond this temperature limit for a longer period of time. The crystallization of the Si-layer can be done either by laser- or by solid-phase crystallization [11].

The Center for Photovoltaic Devices at the University of New South Wales and the Pacific Solar Corp. are exploring the so-called multijunction solar cell concept applying the deposition of a number of layers with alternating p- and n-doping. The basic idea is that very short minority carrier diffusion lengths can be tolerated in these layers and thus very impure material can be used [12].

4.3.2. Si-layers on high-temperature resistant substrates

The high-temperature approaches rely on the high Si-deposition rates ($\cong 5 \mu\text{m}/\text{min}$) at temperatures beyond 1000°C and Si-recrystallization techniques via the liquid phase ($T_{\text{melt, Si}} \sim 1420^\circ\text{C}$). With these technologies, high-throughput, continuous inline-processing, large grains and thus high efficiencies can be obtained. On the other hand, a substrate material with ideal technical properties at a low price could not be found yet.

4.3.3. Transfer technologies of monocrystalline thin Si-films onto glass

The approaches based on transfer technologies of thin monocrystalline Si-films are the latest and very exciting developments within the crystalline Si thin-film community. The basic idea is to detach a semi-processed thin monocrystalline Si-layer from the FZ-wafer and to transfer it onto a glass substrate. The advantages are obvious: the superior material quality enables very high efficiencies even from $<40 \mu\text{m}$ thick Si-layers, inexpensive glass can be used as substrate and finally, the FZ-wafer can be reused for the thin-film formation and detachment for several times. There are currently five main approaches under development [13]: the ψ process [14], the epi-lift process by ANU [15], the so-called VEST-structure (via-hole etching for the separation of the thin film) by Mitsubishi [16]. Sony pursues the epitaxial growth of monocrystalline Si film on the thermally annealed crystalline surface on top of a highly porous buried Si-level. The Si-layer for the solar cell can be detached from the wafer due to the mechanically fragile separation layer underneath [17].

The Institute of Physical Electronics also relies on the epitaxial growth of the active Si-layer on a so-called quasi-monocrystalline Si-film. Again, a buried porous layer enables the separation of the solar cell from the wafer and the transfer of the processed cell to a foreign superstrate. Due to an excellent light-trapping scheme, efficiencies of up to 14.0% were obtained with this approach. Furthermore, it could be shown that the starting wafer can be used several times as a seeding wafer for the porous layer formation and the subsequent epi-growth [18].

5. Amorphous silicon

5.1. History

The first publications on amorphous silicon (a-Si) relevant for solar cell fabrication appeared after the late 1960s [19,20]. The first solar cell from amorphous silicon was reported by Carlson in 1976 [21]. The first consumer products appeared on the market in 1981. The high expectancy in this material was curbed by the relatively low efficiency obtained so far and by the initial light induced degradation for this kind of solar cells (so-called Staebler–Wronski effect [22]). Today, a-Si has its fixed place in consumer applications, mainly for indoor use. After understanding and partly solving the problems of light-induced degradation, amorphous silicon begins to enter the power market. Stabilized cell efficiencies reach 13%. Module efficiencies are in the 6–8% range.

6. a-Si/c-Si heterostructures

A very exciting new development is the combination of crystalline and amorphous technologies in heterostructures. Absorption of sunlight still occurs mainly in a wafer of mono- or poly-crystalline silicon. The silicon wafer is contacted on both sides with amorphous silicon films. This technique represents a combination of c-Si and a-Si technologies.

This configuration has the following advantages: (1) potential for high efficiency, (2) very good surface passivation → low surface recombination velocity, (3) low processing temperatures → all processing steps occur below 200°C, (4) low thermal budget for processing → reduction of energy pay back time, and (5) reduced cost of cell technology.

The best results with this approach were obtained by the Japanese company Sanyo. The latest achievement is a conversion efficiency of 20.7% for a cell area of 101 cm² using n-type CZ-Silicon as the base (light absorbing) part [23].

7. Copper indium diselenide (CIS) and related compounds

7.1. History

Ternary compound semiconductors CuInSe₂, CuGaSe₂, CuInS₂ and their multinary alloy Cu(In,Ga)(S,Se)₂ (abbreviated CIGS) are a very promising thin film technology. Establishing a well-controlled system for multisource coevaporation by Boeing made the CIS cell soon the frontrunner with respect to thin film solar cell efficiencies. ARCO Solar developed in the mid-1980's a fabrication technology which is more adapted to commercial thin film processing, namely sputtering of metal films with a subsequent selenisation step [24]. In later developments the addition of Ga and sulfur helped to increase the efficiency.

7.2. Material properties and deposition techniques

For the large-scale fabrication two different techniques exist for the deposition of the absorber layer:

- deposition of precursor layers and subsequent selenisation by annealing in H₂Se vapor, and
- coevaporation.

The selenisation technique is now the basis of the first pilot production and market introduction of CIS modules by Siemens Solar with efficiencies between 9% and 10%.

The first non-silicon thin film material on the market is CdTe which is already in use in consumer products. Larger production capacities for this material are in the process of being installed in several locations.

8. Tandem cells and materials for concentrating systems

The efficiency of solar cells can be significantly enhanced by stacking cells with different band gaps on top of each other since this makes better use of the energy contained in the solar spectrum. A complication, however, arises from the fact that each subcell has to generate the same electric current in the case of the conventional two-terminal devices. Prominent examples of these tandem or multijunction solar cells are the a-Si/SiGe and the GaInP/GaAs/Ge technologies, the latter being used in space applications. For terrestrial systems the development of an optical concentrator technology is an attractive option with the potential for very low electricity generation cost [25]. Here the required direct sunlight is focussed onto a small-sized solar cell. In a system that concentrates the sunlight by a factor 100–1000, the cost of the solar cell becomes only a minor part of the total system cost. An attractive feature of concentration is that the solar cell conversion efficiency increases with the light intensity by something like 10–20% relative for the above-mentioned concentrations, provided the cell temperature is kept constant. Concentrator cell and module efficiencies above 30% and 25%, respectively, have already been demonstrated. Market deployment of concentrator systems has so far proved to be slow because of a comparatively small market for the appropriate medium-sized (100 kW) remote PV power systems. Another issue is the largely unproven reliability of the concentrator technology. An obvious and logical development path is therefore to pursue the available crystalline silicon-based point-focus systems in order to continue developing concentrator system technology and markets and then make the transition to the more efficient III–V-based multijunction cells as these become more and more mature.

9. Dye-sensitized cells

Nano-crystalline dye-sensitized solar cells are based on the mechanism of a fast regenerative photoelectrochemical process [26]. The main difference of this type of solar cells compared to conventional cells is that the functional element, which is responsible for light absorption (the dye), is separated from the charge carrier transport itself. In the case of the n-type semiconductor TiO_2 (band gap 3.2 eV), this results in a working cycle starting with the dye excitation by an absorbed photon at the TiO_2 /electrolyte interface and an electron injection into the TiO_2 . The injected electrons may migrate to the front electrode (a transparent TCO glass) and can be extracted as an external current. The dye is subsequently reduced by a redox electrolyte, based on an organic solvent and the redox couple iodide/triiodide.

The major advantage of the concept of dye sensitization is the fact that the conduction mechanism is based on a majority carrier transport as opposed to the minority carrier transport of conventional inorganic cells. Thus, impure starting materials and a simple cell processing without any clean room steps are permitted, yet resulting in promising conversion efficiencies of 7–11% and the hope of a low-cost device for photoelectrochemical solar energy conversion. The most important

issue of the dye-sensitized cells is the stability over the time and the temperature range which occurs under outdoor conditions.

10. Organic solar cells

Besides dye-sensitized solar cells, which may be considered as organic/inorganic hybrid cells, other types of organic solar cells have currently become of broader interest. These cells can be divided roughly into molecular and polymer organic solar cells or into flat-layer systems and bulk heterojunctions.

Extremely high optical absorption coefficients are possible with these materials, which offers the possibility for the production of very thin solar cells (far below $1\text{ }\mu\text{m}$) and therefore only very small amounts of needed materials. Considering the fact that light-emitting films of plastic materials have been realized there is a chance to achieve photovoltaic conversion also in such materials.

Only modest solar conversion efficiencies of up to 1% were reached until 1999 but efficiencies then increased rapidly. With molecular flat-layer systems based on molecular organic single crystals made of iodine- or bromine-doped pentacene, efficiencies of up to 3.3% under AM1.5 illumination have been reported at Lucent Technologies. Nearly the same value was reported with improved bulk heterojunctions (interpenetrating network) of conjugated polymers and fullerene derivatives [27].

Before these cells become practical, which at the moment looks still far away, the efficiency will have to be increased further. Also, long-term stability and protection against environmental influences are significant challenges.

11. Theoretical concepts for new high-efficiency semiconductor materials

This chapter deals with new theoretical materials that do not yet exist but would offer significant advantages for photovoltaic conversion. They might be called designer materials. In principle they are designed to combine the effect of tandem cells in one material.

11.1. Auger generation material

In this concept the higher energy photons (energy greater than $2E_G$) should generate two or even more electron hole pairs by impact ionization [28,29]. As a possible material a Si-Ge alloy has been identified.

11.2. Intermediate metallic band material

The intermediate metallic band material solves a problem that has a long history in solar cell materials design. Photons with energy less than the gap could be utilized if an intermediate energy level around midgap were present through which carriers

could be transported from one band edge to the other by two photons. Unfortunately such levels are also strong recombination centers and lead to drastic degradation of the material. This problem is circumvented (at least theoretically) by placing a narrow metallic band within the gap of a wide gap semiconductor [30]. The intermediate band prevents nonradiative recombination. The theoretical efficiency of this device is 46.0%, compared to 41.9% of the tandem cell described above. A proposal already exists how to manufacture the intermediate band material. It is proposed to incorporate quantum dots of 35–70 Å diameter of a suitable ternary compound into another, wider gap ternary material [31]. Whether this material can be synthesized, remains an open question.

12. Future development of solar cell efficiency

An attempt will now be made to predict the future development of solar cell efficiency. It can be observed that every solar cell technology follows a certain pattern in its development. At the inception of a new technology efficiency increases rapidly, after some time the development slows down, approaching a limiting efficiency that cannot be exceeded. This applies in particular to the best laboratory efficiencies which will only be considered here. If a function can be found describing the past development it can be expected that it can be extrapolated into the future with some confidence.

The function chosen here (Eq. (1)) is relatively simple (it is derived from the diode equation):

$$\eta(t) = \eta_L(1 - \exp((a_0 - a)/c)), \quad (1)$$

where $\eta(t)$ is the time-dependent efficiency, η_L is the limiting asymptotic maximum efficiency, a_0 is the year for which $\eta(t)$ is zero, a is the calendar year and c is a characteristic development time. (The choice of function is of course not exclusive.) The function can be fitted relatively well to the past development of best laboratory efficiencies of crystalline silicon cells. We now can predict how this efficiency will develop in the future. The limiting efficiency from the fit is 29% but this cannot be

Table 1
Fitting parameters for different technologies

Technology	η_L	c	a_0
Xtal Si	29	30	1948
Thin film Si	30	19	1989
CIS/CIGS	29	30	1969
a-Si	18	20	1968
Organic cells	18	25	1995
New material	42	25	2000

Thin film silicon: very few data points; a-Si: large scatter of data; organic cells: only one data point, undetermined parameters were estimated.

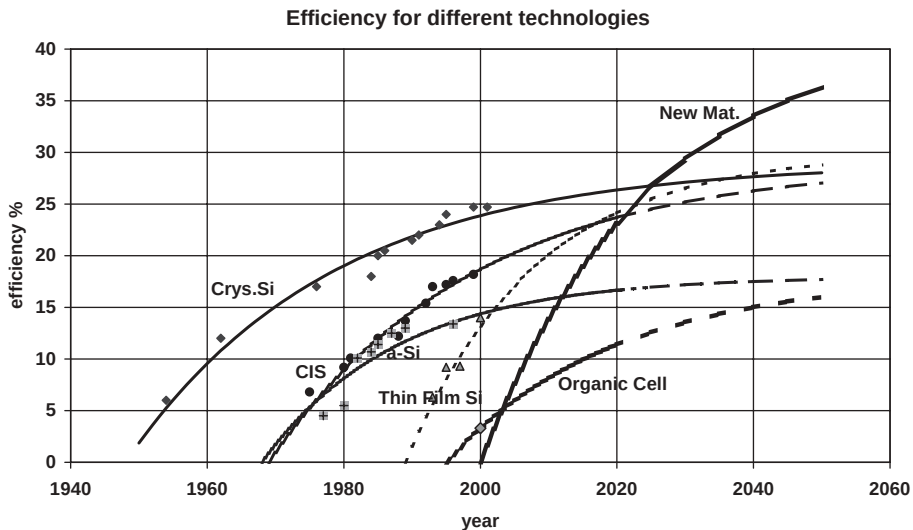


Fig. 1. A century of photovoltaics. Past achievements and future predictions of highest laboratory solar cell efficiencies. Best data are for crystalline silicon. The right curves are based on very few data points, the curve for a new material is purely hypothetical.

reached in finite time. Fitting of the curve is done with three parameters which are given in Table 1 for silicon and the most important other technologies (Fig. 1). The curve to the right describes a hypothetical new high efficiency material or combination of materials, the development of which is (optimistically) assumed to be starting now.

All the curves in their long-term perspective spanning a century of past and future development are shown in Fig. 1. For the curves in Fig. 1 to become reality a precondition is that somebody is working on those developments. This is unlikely in the asymptotic part of the curves because a large effort is required for a very small improvement.

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